习题解答

4-4： $2\times[2\times(3)-(2)-(1)]$ 得：

$$\Delta_r G_m^{\ominus} = 4\times\Delta_r G_m^{\ominus}(3) - 2\times\Delta_r G_m^{\ominus}(2) - 2\times\Delta_r G_m^{\ominus}(1)$$

$$K^{\ominus}(4) = \left\{\frac{[K^{\ominus}(3)]^2}{K^{\ominus}(1)\cdot K^{\ominus}(2)}\right\}^2$$

4-5： $2\times(2)-2\times(1)$ 得：

$$K^{\ominus} = [K^{\ominus}(2)]^2/[K^{\ominus}(1)]^2 = (1.8\times10^4)^2/0.80^2 = 5.06\times10^8$$

4-9： 解：

	$N_2O_4(g)$	=	$2NO_2(g)$
起始：	n		0
平衡：	$n-n\alpha$		$2n\alpha$

$n_{总} = n + n\alpha$　　$x(N_2O_4) = (1-\alpha)/(1+\alpha)$　　$x(NO_2) = 2\alpha/(1+\alpha)$

习题解答

$p(N_2O_4) = [(1-\alpha)/(1+\alpha)]p_{总}$　　$p(NO_2) = [2\alpha/(1+\alpha)]p_{总}$

$$K^{\ominus} = \frac{[p(NO_2)/p^{\ominus}]^2}{p(N_2O_4)/p^{\ominus}} = \frac{[\frac{2\alpha}{1+\alpha} \times 400/100]^2}{\frac{1-\alpha}{1+\alpha} \times 400/100} = 1.00$$

$$\alpha = 24.25\%$$

当 $p_{总} = 1000$ kPa 时，同理代入得 $\alpha' = 15.62\%$，说明增大压强反应向分子数减少的方向移动。

习题解答

4-10：解：　$NH_4Cl\,(s) = NH_3(g) + HCl\,(g)$

$\Delta_f H_m^\ominus$　　–314.4　　–46.11　　–92.31

$S_m^\ominus$　　94.56　　192.3　　186.8

$\Delta_r H_m^\ominus = \sum \nu_B \Delta_f H_m^\ominus (B) = 175.98\ kJ\cdot mol^{-1}$

$\Delta_r S_m^\ominus = \sum \nu_B S_m^\ominus (B) = 284.54\ J\cdot mol^{-1}\cdot K^{-1}$

$\Delta_r G_m^\ominus(298K) = \Delta_r H_m^\ominus (298K) - T\,\Delta_r S_m^\ominus (298K)$

$= 175.98\times10^3 - 298\times284.54$

$= 91200\ J\cdot mol^{-1} > 0$　不自发！

$\Delta_r G_m^\ominus(298K) = -RT\ln K^\ominus$　　$K^\ominus = 1.06\times10^{-16}$

$\Delta_r G_m^\ominus(298K) = \Delta_r H_m^\ominus (298K) - T\,\Delta_r S_m^\ominus (298K) < 0$

$175.98\times10^3 - T\times284.54 < 0$

$T > 615\ K$

习题解答

4-11：解：

	C (石墨)	+ CO_2 (g)	= 2CO (g)
$\Delta_f H_m^\ominus$	0	–393.51	–110.52
$S_m^\ominus$	5.74	213.6	197.56

(1) $\Delta_r H_m^\ominus = \sum \nu_B \Delta_f H_m^\ominus (B) = 172.47\ kJ\cdot mol^{-1}$

$\Delta_r S_m^\ominus = \sum \nu_B S_m^\ominus (B) = 175.78\ J\cdot mol^{-1}\cdot K^{-1}$

$\Delta_r G_m^\ominus(298K) = \Delta_r H_m^\ominus (298K) - T\,\Delta_r S_m^\ominus (298K)$

$= 172.47\times10^3 - 298\times175.78$

$= 120000\ J\cdot mol^{-1} > 0$ 不自发！

$\Delta_r G_m^\ominus(298K) = -RT\ln K^\ominus(298K)$ $K^\ominus(298K) = 9.23\times10^{-22}$

(2) $\Delta_r G_m^\ominus(1198K) = \Delta_r H_m^\ominus (298K) - T\,\Delta_r S_m^\ominus (298K)$

$= 172.47\times10^3 - 1198\times175.78$

$= -38114\ J\cdot mol^{-1}$

$$\Delta_r G_m^{\ominus}(1198\text{K}) = -RT\ln K^{\ominus}(1198\text{K}) \qquad K^{\ominus}(1198\text{K}) = 45.91$$

(3) $\Delta_r G_m(1198\text{K}) = \Delta_r G_m^{\ominus}(1198\text{K}) + RT\ln Q$

$$= -38114 + 8.314\times1198\ln[(p(\text{CO})/p^{\ominus})^2/(p(\text{CO}_2)/p^{\ominus})]$$

$$= -38114 + 8.314\times1198\ln[(100/100)^2/(200/100)]$$

$= -45016\ \text{J}\cdot\text{mol}^{-1} < 0$ 自发进行！

4-12：解法一

$$\ln\frac{K_2^{\ominus}}{K_1^{\ominus}} = \frac{-\Delta_r H_m^{\ominus}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln\frac{K_2^{\ominus}}{9.07} = \frac{-(-37.9\times10^3)}{8.314}\left(\frac{1}{800} - \frac{1}{700}\right)$$

$$K_2^{\ominus} = 4.02$$

习题解答

4-11：解法二　　700 K 时：

$$\Delta_r G_m^{\ominus}(700\text{K}) = -RT\ln K^{\ominus}(700\text{K}) = \Delta_r H_m^{\ominus} - T\Delta_r S_m^{\ominus}$$

$$-8.314\times700\times\ln 9.07 = -37.9\times10^3 - 700\Delta_r S_m^{\ominus}$$

$$\Delta_r S_m^{\ominus} = -35.81\ \text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$$

$$\Delta_r G_m^{\ominus}(800\text{K}) = \Delta_r H_m^{\ominus} - T\Delta_r S_m^{\ominus}$$

$$= -37.9\times10^3 - 800\times(-35.81)$$

$$= -9252\ \text{J}\cdot\text{mol}^{-1}$$

$$\Delta_r G_m^{\ominus}(800\text{K}) = -RT\ln K^{\ominus}(800\text{K})$$

$$= -8.314\times800\times\ln K^{\ominus}(800\text{K})$$

$$K^{\ominus}(800\text{K}) = 4.02$$